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# Understanding AI Buzzwords LLM, RAG, AI Agents & MCP — Made Simple

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*Demystifying the jargon flooding your feed*

# Understanding AI Buzzwords LLM, RAG, AI Agents & MCP Made Simple 🤔

→ Demystifying the jargon flooding your feed



AI is everywhere. Whether you're scrolling through social media, reading blog posts, or exploring tutorials, you've probably seen terms like **LLM**, **RAG**, and **AI Agents**. But what do they really mean ?

Let's break them down with simple examples.

## What is an LLM?

LLM = Large Language Model

Think of it as a supercharged autocomplete. It reads massive amounts of text (books, websites, conversations) and learns to guess the next word in a sentence.

*Example:*

You type: “*Once upon a...*”

It completes: “*...time, there was a bird who loved to code.*”

It doesn't understand like a human, but it's great at mimicking patterns in language.

## What is an AI Assistant ?

AI Assistant = LLM + User Interface

An **AI assistant** is a friendly interface powered by an **LLM**. It can chat, explain concepts, summarize content, or even write emails.

DeepSeek, ChatGPT, Gemini, Mistral & Apple Intelligence are AI Assistants.

You ask: “*Explain Blockchain like I'm 10.*”

It simplifies it thanks to its **language abilities**.

Sometimes assistants use **SLMs** (Small Language Models) when **speed, privacy**, or **low-power** use is important

*Example :*

Apple Intelligence uses an **on-device SLM** to handle simple offline tasks like text summarization.

👉 See my full presentation for a deep dive into [Apple Intelligence architecture](#).

## What is RAG ?

LLMs are great at guessing, but *they don't “know” anything*. They don't search the web or read your files or database. So when they don't know, they might:

- Hallucinate (make stuff up) 🤖
- Say “I don’t know” 🙋

### 💡 **RAG to the rescue !**

RAG = Retrieval-Augmented Generation

Here’s what happens:

1. **Retrieve** : Finds relevant content from a *database, internet, or document*
2. **Augment** : Adds it to the LLM’s input
3. **Generate** : Produces a better, grounded response

With RAG, you can build **smarter** and **safer** assistants based **on your data**.

## What are AI Agents ?

AI agents don’t just answer questions, **they take action**.



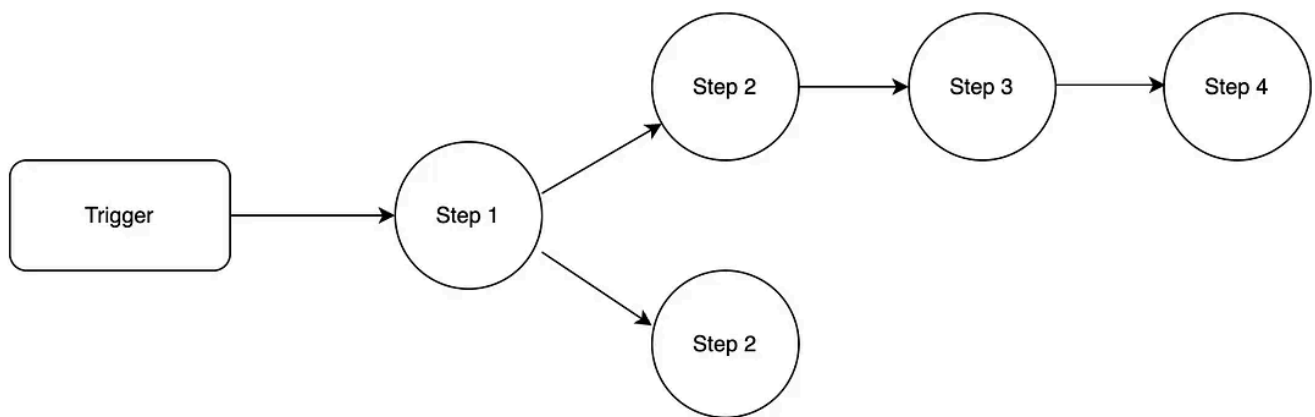
How it works :

1. You give it a **goal (input)** : “Book a flight and hotel for a Tunis conference.”
2. It **plans** : figures out **steps** like searching, filtering and comparing
3. It uses tools : **APIs, calendars, scripts, emails**
4. It **acts (output)** : books, confirms, updates you

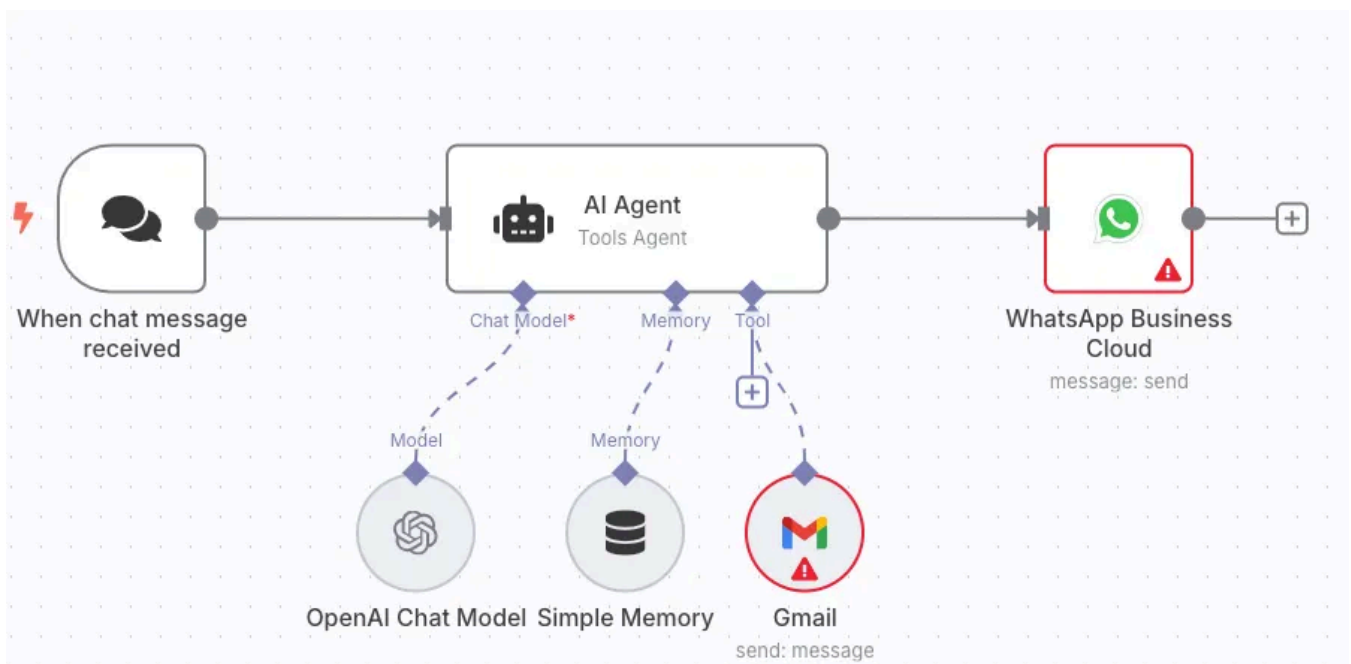
**Agents vs Traditional Automation :**

- Automation workflows = static user **pre-defined steps** (like a checklist)

- AI agents = the agent make the **decision and the orchestration** for you based on your **inputs** (a smart assistant)



Traditional Automated Workflow



AI Agent Workflow

➡ “Agents are flexible. Workflows are rigid.”

## What is MCP (Model Context Protocol) ?

MCP = Structured context for intelligent agents

Designing scalable AI agents requires powerful **context management**.

To build distributed systems where multiple APIs, tools, and LLMs collaborate, we need a standard way to **structure** and **share context**.

⚡ MCP is a standard developed and **open-sourced** by Anthropic to provide this structure.

*“MCP is to AI agent architecture what REST APIs were to the web architecture.”*

## What is A2A Protocol ?

A2A (Agent-to-Agent) is an open-source protocol by Google designed to enable direct **communication between autonomous agents**.

Unlike the **Model Context Protocol (MCP)** which focuses on standardizing tool use for **single agents**, A2A expands the scope to **agent collaboration**, and coordination.

⚡ MCP is a **lightweight protocol** focused on **tool invocation** for LLMs.

→ Think: “one agent, many tools.”

⚡ A2A is a **coordination protocol** that lets **multiple agents communicate, collaborate, and delegate**.

→ Think: “many agents, shared goals.”

➡ You don't have to choose one or the other : An agent can use MCP to call tools and use A2A to talk with other agents

# MCP vs. A2A

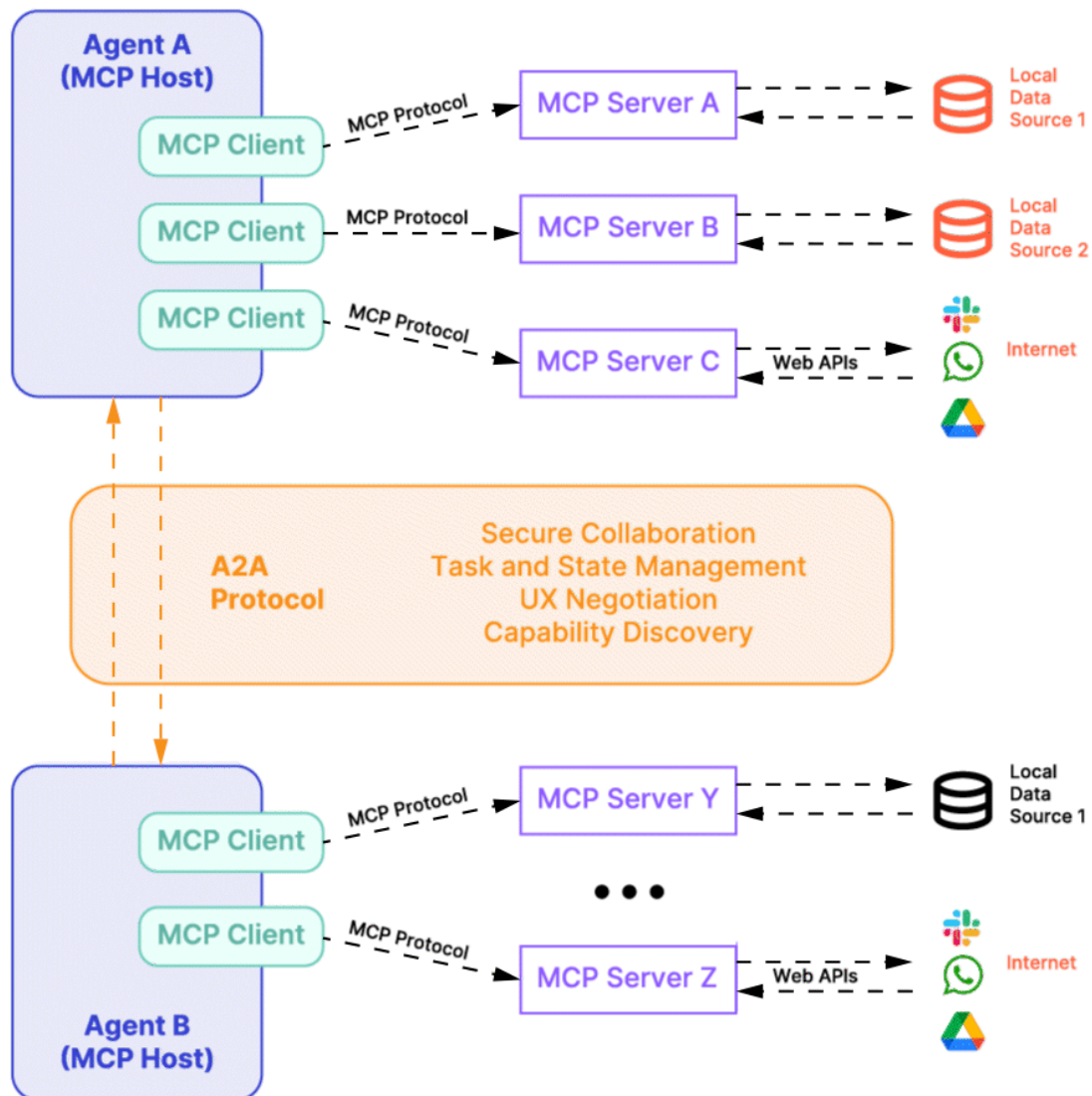


Image by [Generative AI](#)

## Now putting it all together :

LLMs help us generate language ✅

RAG gives them real knowledge ✅

AI Agents help them make decisions and act ✅

MCP & A2A help them think together in context 

## What about tools like n8n, Make, or Zapier ?

The good news ? 

Low-code platforms like Zapier, Make, and n8n make it easy to build smart assistants by orchestrating LLMs, tools, APIs, and **context** through visual workflows, so no need for deep technical knowledge to get started.

Using one of these tools you can :

- **Trigger** AI actions based on **events** (e.g. “New email → summarize it”, “Merge request” → “Review it”)
- **Chain tools** and **LLMs** together in workflows
- **Connect** to APIs, databases, and document sources
- Add **memory** or **context** manually or via external services

Even if they don't fully implement protocols like MCP, you can still build **MCP-inspired logic**: share **memory** between steps, route data, and manage smart assistants across your systems.

 In the next article, we'll walk you through a step by step tutorial to show you how to build an AI agent that can **review your code** , understand **pull request**, and give suggestions like a real teammate .

## Thank you for being a part of the community

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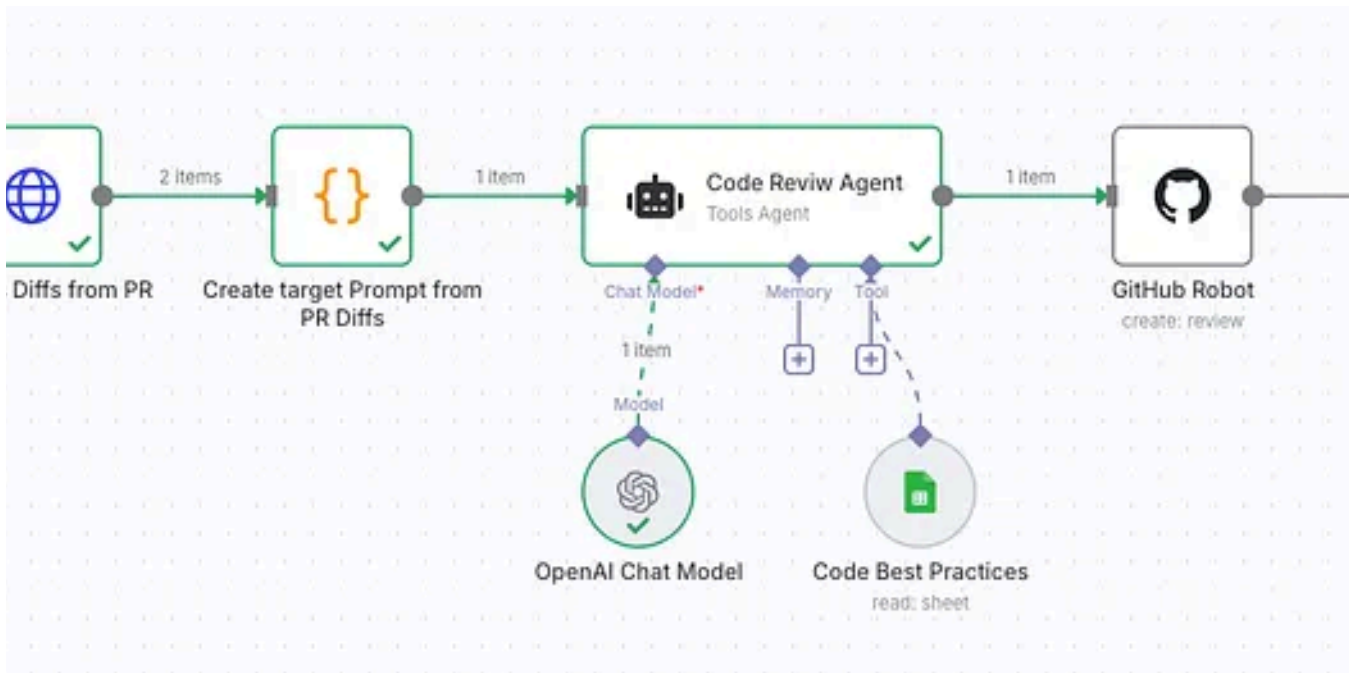


Very well explained. Thank you!!



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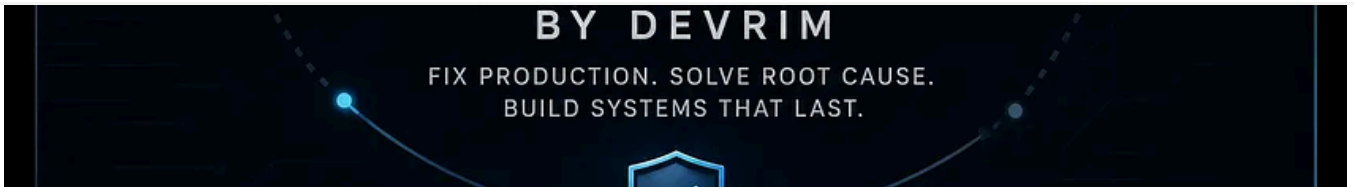
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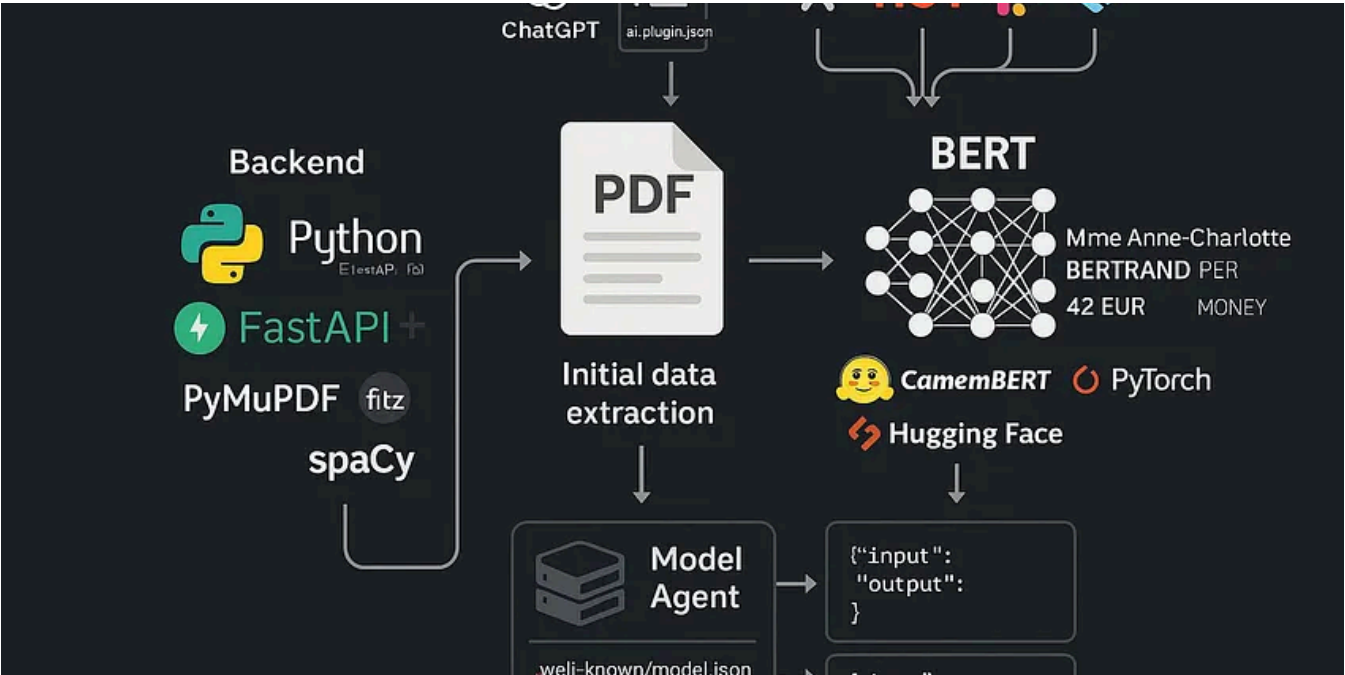


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